V60 Diode Laser spectroscopy

Abstract

In this experiment the characteristics of a tunable diode laser and the operation mode of a internal and an external cavity gets to know. In the second step the laser is used to measure the energy splitting of Rb-atoms with the technique of absorption spectroscopy.

References

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[3] H. J. Eichler, J. Eichler and O. Lux, *Lasers (Basics, Advances and Application)*, Springer (2018)

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Preparation

To do the experiment it is necessary to answer the following guiding questions.

- What is coherent light? How do you get population inversion? How does light amplification in p-n diodes work?
- What are the difference in electronic structure between metals, semiconductors and isolators. Explain it with the help of valence and conduction bands.
- What is a p- and n-doped semiconductor?
- Discuss the working principle of a diode laser?
- Discuss the internal and the external cavity of a diode-laser. What is a Littrow configuration?
- Which laser modes can be get? Explain the concept of ´mode hopping´. What defines the wavelength of a diode laser?
- To which class of materials does rubidium belong? What splittings are caused by an external magnetic field? Calculate the hyperfine splitting of rubidium.What are the selection rules for magnetic diple transistions?
- What does absorption spectroscopy mean?

Experimental setup and Alignment

Main part of the experiment is a diode laser head including the diode laser, a collimating lense, a thermo-electric heating and a diffraction grating. The diode is based on a p-n semiconductor chip made from aluminium gallium arsenide (AlGaAs) and have a emission wavelength of about 780 nm. The chip itself build a resonator with 100 % reflectivity one one side of the chip and 15 % reflectivity on the other side. A lense is necessary to collimate the beam. A diffraction grating in Littrow configuration is used as external cavity to stabilize the laser at a defined wavelength. Together with a temperature change, the wavelength can be variied. Current, temperature and the grating angle are controlled with TeachSpin´s *laser diode controller*.

The emission wavelength of the diode laser is in the near infrared, so that it is necessary

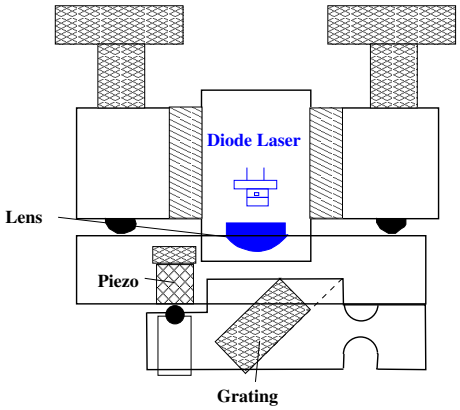


Figure 1: Sketch of the laser head

to use an indication card, a photodiode or a CCD-Camera to ´see´ the laser emission. To measure the threshold current place a piece of paper in front of the laser head and detect the reflected light with the CCD Camera. Increase the laser current while watching the monitor of the CCD. At the threshold current suddenly the brightness of the beam spot increase and you see a speckle pattern which is characteristic for lasing.

To align the external cavity you must adjust the angular orientation of the grating so that the second order diffraction from the grating is going back to the diode. This is done by adjusting the grating in such a way that the primary order spot and the secondary order spot overlaps at the monitor. Replace the CCD Camera with a photodiode and connect a voltmeter to the monitor output. By changing the lens position you will get the maximum output power.

To observe the fluoresence of the hyperfine splitting of rubidium a rubidium assambly can place into the laser beam. The rubidium cell assambly contains the rubidium cell, filled with ^{85}Rb (72 % abundance) and ^{87}Rb (28 % abundance), a heater, a cooling finger and a pair of helmholtz coils. Each coil have a radius of 87.4 mm and 320 turns of copper wire. To observe the absorption spectrum of rubidium use the experimental configuration of Figure 2. The two photodiodes are used to reduce noise.

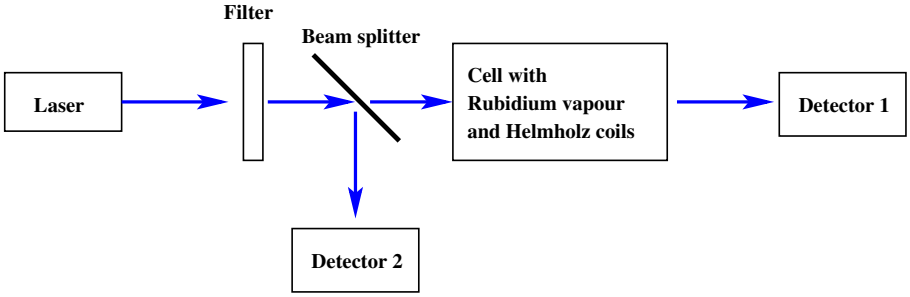


Figure 2: Experimental setup.

Measurements and analysis

- Study the characteristics of the diode laser by measuring the threshold current for lasing. Observe mode hopping of the diode laser and document it with an oscilloscope. Change the external cavity length and measure its effect.
- Measure the hyperfine splitting of ^{85}Rb and ^{87}Rb using absorption spectroscopy. To do this measure the transmittet light as a function of the laser frequency. Identify and measure the hyperfine splitting of Rb. Compare the theoretical relation of ^{85}Rb and ^{87}Rb with the experimental one.